

Derbyshire Shared Care Pathology Guidelines

Hypocalcaemia in Adults

Purpose of Guideline

The investigation and management of patients with newly diagnosed hypocalcaemia

Definition

Adjusted (corrected) calcium <2.20 mmol/L

Corrected or Adjusted Calcium

We report total calcium and corrected (adjusted for albumin) calcium results.

All the calcium values in this guideline refer to the corrected values.

This adjustment is only an approximation.

The correction is not valid in acid-base disorders. For example although the corrected calcium may be normal in acute respiratory alkalosis, the ionised calcium may be low, and the patient may have symptoms of hypocalcaemia, such as parasthesia.

What are the common signs / symptoms?

- Mild hypocalcaemia is not uncommon, and in many patients the only symptoms may be vague muscular aches and pains
- The number and severity of symptoms will be dependent on the degree of hypocalcaemia
- Severe symptoms such as tetany are rare, but parasthesia can be seen. Cardiovascular complications, arrhythmias, hypotension and heart failure can all be seen.

What are the causes of hypocalcaemia?

- Actual Vitamin D Deficiency
 - Dietary
 - Lack of Sunlight
 - Malabsorption, especially pancreatic disease and coeliac disease
- Functional Vitamin D Deficiency
 - Renal Disease (lack of 1-Hydroxylation)
 - Liver Disease (lack of 25-Hydroxylation)
- Magnesium deficiency
- Hypoparathyroidism
 - Autoimmune
 - Post-surgery

When is hypocalcaemia considered a medical emergency?

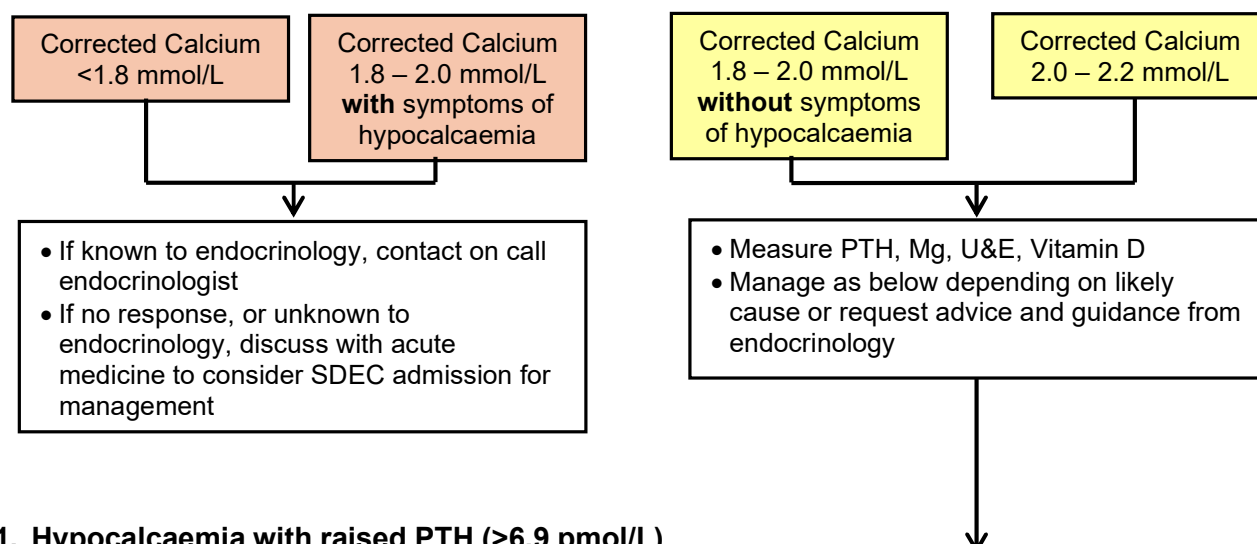
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|--------------------|--------------------|-----------------------------|
| • Adjusted Calcium | 2.00 – 2.20 mmol/L | Not a medical emergency |
| • Adjusted Calcium | 1.80 – 2.00 mmol/L | Possible medical emergency |
| • Adjusted Calcium | <1.80 mmol/L | Usually a medical emergency |

These limits are for guidance only; the severity of the patient's symptoms, and other factors, e.g. rapidity of the fall in calcium, renal function, will also determine whether the patient requires emergency admission.



- Calcium results ≤ 1.80 mmol/L will usually be telephoned to Derbyshire Health United when the GP practice is closed

Management of Hypocalcaemia



1. Hypocalcaemia with raised PTH (>6.9 pmol/L)

PTH	Vit D	Mg	eGFR	Most likely cause
High	<30 nmol/L	>0.4 mmol/L	>30	Vitamin D deficiency Treat with rapid oral Vitamin D replacement: see JAPC Vitamin D replacement guideline http://www.derbyshiremedicinesmanagement.nhs.uk/clinical-guidelines/chapter_9/
High	>30 nmol/L	>0.4 mmol/L	<30	Active Vitamin D (1,25-OH Vit D) deficiency Requires Renal opinion

- Vitamin D deficiency is the usual diagnosis of hypocalcaemia with raised PTH
- If the patient's renal function is normal the patient can present with a low serum phosphate
- Treatment can commence straight away
- If renal, gastrointestinal or hepatic involvement is present the appropriate referral should be made when required

2. Hypocalcaemia with normal or decreased PTH (<6.9 pmol/L)

PTH	Vit D	Mg	eGFR	Most likely cause
Normal or Low	>30 nmol/L	<0.7 mmol/L	>30	Magnesium deficiency If on PPI consider stopping. See Shared Care Pathology hypomagnesaemia guideline for advice on replacement: • Mg 0.4 - 0.7 mmol/L = oral replacement • Mg <0.4 mmol/L = i.v. replacement, i.e. admit patient same day or discuss with endocrinologist https://www.uhdb.nhs.uk/download.cfm?doc=docm93jijm4n854.pdf&ver=44148
Normal or Low	>30 nmol/L	>0.7 mmol/L	>30	Hypoparathyroidism Requires Endocrine opinion

- In patients with low serum calcium and a PTH within the reference range it is important to rule out magnesium deficiency, which gives rise to a blunted PTH response to hypocalcaemia
- Magnesium deficiency often requires correction to enable the calcium to be normalised
- In patients with low serum calcium, a PTH below the reference range is virtually diagnostic for hypoparathyroidism

Causes of Magnesium Deficiency

Renal Loss

- Diuretics, especially Loop Diuretics
- Cytotoxic Drugs
- Aminoglycosides
- Immunosuppressants
- Drugs, e.g. PPI's, Bisphosphonates

GI

- Severe Diarrhoea
- Malnutrition
- Alcoholism

Magnesium Deficiency: Shared Care Pathology guideline

<https://www.uhdb.nhs.uk/download.cfm?doc=docm93jjm4n854.pdf&ver=44148>

Vitamin D Deficiency: JAPC guideline

http://www.derbyshiremedicinesmanagement.nhs.uk/clinical-guidelines/chapter_9/

Contacts

UHDB

Duty Biochemist	01332 789383 (8am to 7pm, Mon – Fri)
On Call Consultant Biochemist	Via RDH switchboard, 01332 340131 (24/7)
Endocrinology Advice	07879 115507 (9am – 5pm, Mon – Fri)
Renal Registrar (9am to 9pm)	via RDH switchboard bleep 8121
GP Phone Triage	01332 789099
MAU and ACC	01332 788707 <u>OR</u>
MAU Nurse in Charge	07917 650751

Chesterfield

Duty Biochemist	01246 512212 (9am to 5pm, Mon – Fri)
On Call Consultant Biochemist	Via CRH switchboard, 01246 277271
A&E, EMU and CDU	Via CRH switchboard

Burton

Duty Biochemist	01283 566333 ext 4233
Endocrinology advice	Via endocrine secretaries 01283 593259

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Dr D Hughes, Dr R Robinson, Dr P Blackwell, Mrs H Seddon	Jan 2022	31 st Jan 2024
Dr R Stanworth, Dr P Blackwell, Mrs H Seddon	Apr 2024	30 th Apr 2026